

# Bank Interest Program

## Aim/Objective:

To develop a Java program to demonstrate method overriding and runtime polymorphism by finding the rate of interest for different banks.

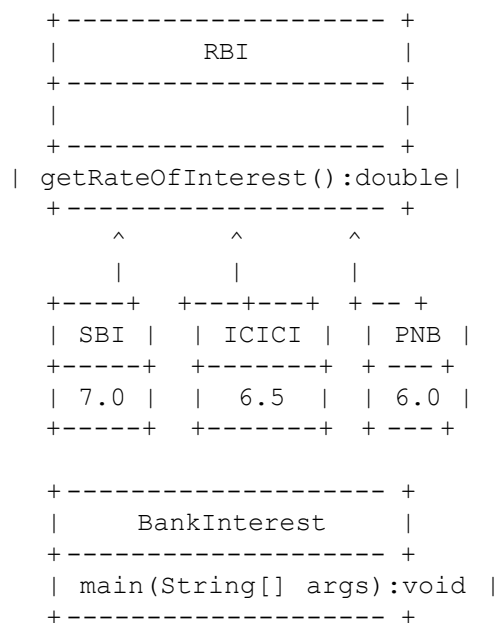
## System Requirements:

- Operating System: Windows or any operating system supporting Java.
- Programming Language: Java.
- Java Development Kit (JDK).
- Text Editor or IDE: VS Code / Eclipse / IntelliJ IDEA.
- Command Prompt or Terminal.

## Algorithm/Logic Description:

1. Start.
2. Create a base class named RBI with a method `getRateOfInterest()`.
3. Create SBI, ICICI, and PNB classes by extending RBI.
4. Override `getRateOfInterest()` in each bank class with its respective interest rate.
5. Read the bank name from the user.
6. Compare the entered bank name using `equalsIgnoreCase()`.
7. Create the corresponding bank object and assign it to an RBI reference variable.
8. If the bank name is invalid, display an invalid bank message and stop.
9. Call `getRateOfInterest()` using the RBI reference.
10. Display the interest rate of the selected bank.
11. Close the Scanner.
12. Stop.

## UML Class Diagram:



## Source code

```
class SBI extends RBI
{ @Override
double
    getRateOfInterest(
    ) { return 7.0;
    }
}

class ICICI extends RBI
{ @Override
double
    getRateOfInterest(
    ) { return 6.5;
    }
}

class PNB extends RBI
{ @Override
double
    getRateOfInterest(
    ) { return 6.0;
    }
}

public class BankInterest {
    public static void

        main(String[] args) {

            Scanner sc = new

            Scanner(System.in);

            System.out.println("Enter the Bank name to find the rate
            of Interest:"); String bank = sc.nextLine();

            RBI rbi;

            if
                (bank.equalsIgnoreCase("
                RBI")) { rbi = new
                RBI();
            }
            else if
                (bank.equalsIgnoreCase("SBI"
                )) { rbi = new SBI();
            }
            else if
                (bank.equalsIgnoreCase("ICIC
                I")) { rbi = new ICICI();
            }
            else if
                (bank.equalsIgnoreCase("PNB"
                )) { rbi = new PNB();
            }
            else {
                System.out.println("Invalid Bank
                Name"); sc.close();
                return;
            }
        }
    }
}
```

```
        System.out.println("RBI rate of interest is : "
                            + rbi.getRateOfInterest() + "%");

        sc.close();
    }
}
```

### Compilation and Execution Commands:

Save the file as: BankInterest.java

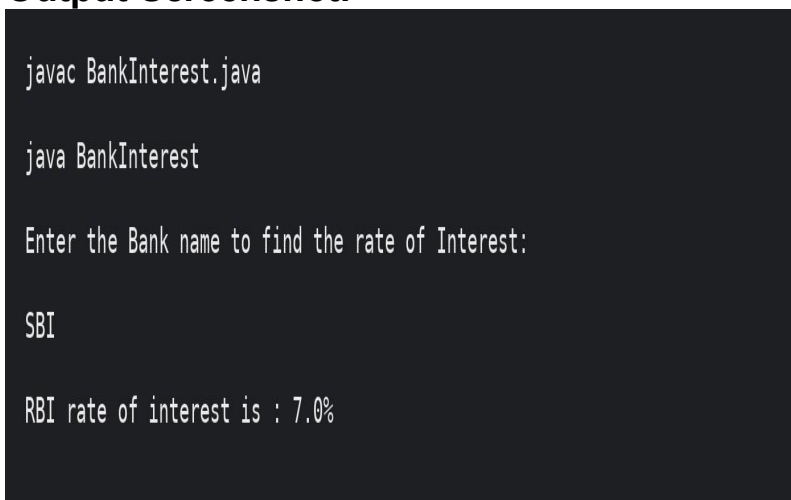
Compile:

```
javac BankInterest.java
```

Run:

```
java BankInterest
```

### Output Screenshot:



```
javac BankInterest.java

java BankInterest

Enter the Bank name to find the rate of Interest:

SBI

RBI rate of interest is : 7.0%
```

### Result:

Thus, the Java program was successfully executed and method overriding with runtime polymorphism was demonstrated by displaying the rate of interest for the selected bank.

